

RMO– 2017

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let $\angle AOB$ be given angle less than 180° and let P be an interior point of the angular region determined by $\angle AOB$. Show, with proof, how to construct, using only ruler and compasses, a line segment CD passing through P such that C lies on the ray OA and D lies on the ray OB , and $CP:PD = 1:2$.
2. Show that the equation:

$$a^3 + (a+1)^3 + (a+2)^3 + (a+3)^3 + (a+4)^3 + (a+5)^3 = b^4 + (b+1)^4$$
has no solution in integers a, b .
3. Let $P(x) = x^2 + \frac{1}{2}x + b$ and $Q(x) = x^2 + cx + d$ be two polynomials with real coefficients such that $P(x)Q(x) = Q(P(x))$ for all real x . Find all the real roots of $P(Q(x)) = 0$.
4. Consider n^2 unit squares in the xy -plane centered at point (i, j) with integer coordinates, $1 \leq i \leq n, 1 \leq j \leq n$. It is required to color each unit square in such a way that when ever $1 \leq i < j \leq n$, and $1 \leq k < l \leq n$, the three squares with centres $(i, k), (j, k), (j, l)$ have distinct colours. What is the possible number of colour needed?
5. Let Ω be a circle with a chord AB which is not a diameter. Let Γ_1 be a circle on one side of AB such that it is tangent to AB at C and internally tangent to Ω at D . Likewise, let Γ_2 be a circle on other side of AB such that it is tangent to AB at E and internally tangent to Ω at F . Suppose the line DC intersects Ω at $X \neq D$ and the line FE intersects Ω at $Y \neq F$. Prove that XY is a diameter of Ω .
6. Let x, y, z be real numbers, each greater than 1. Prove that:

$$\frac{x+1}{y+1} + \frac{y+1}{z+1} + \frac{z+1}{x+1} \leq \frac{x-1}{y-1} + \frac{y-1}{z-1} + \frac{z-1}{x-1}$$

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